

DE Lab Questions

1.

A. Implement Basic Data Handling Commands:

1. Read data from the CSV file

```
import pandas as pd

df=pd.read_csv(r"C:\Users\DELL\Downloads\data1.csv (2).csv")

df
```

2. Dimension of the data

```
df.shape
```

3. Display data (top 5 rows and total data) (Choose your own dataset)

```
print("Top 5 rows:\n", df.head())
print("Total data:\n", df)
```

B. Create a partition (shopping3) for table shopping1 and also create 3 buckets inside each partition, check your partition.

1. Open terminal in the vm
2. Type hive
3. create database shopping;
4. use shopping;
5. create table shopping1(
code int,
item_name string,
category string,
place string
)

row format delimited fields terminated by ',';

in a new terminal type

6. gedit shop.txt

in that type randome stuff like:

```
1,purse,bag,shimla
2,lipstick,cosmetic,delhi
3,bowl,utensils,jammu
4,mobile,electronicgadget,hyderabad
5,skirt,apparel,chennai
6,bedcover,furnishing,chandigarh
7,car,toys,karnal
8,handpurse,bag,solan
9,cream,cosmetic,jodhpur
10,plate,utensils,Mohali
```

And then save it

Get back to the hive terminal

```
7. LOAD DATA LOCAL INPATH '/home/cloudera/Desktop/shop.txt'
    OVERWRITE INTO TABLE shopping1;
```

If errors come remove overwrite

Note: incase it says file missing or smtg then the path must be wrong

In a different terminal **type find /home -name "shop.txt"** to get correct path.

Now partiotioning+bucketing

8.

```
set hive.enforce.bucketing=true;
```

```
create table shopping3(
code int,
item_name string,
place string
)
partitioned by (category string)
clustered by (place) into 3 buckets
row format delimited fields terminated by ',';
```

9.

```
-- Load data
from shopping1
insert overwrite table shopping3 partition(category)
select code, item_name, place, category
distribute by category;
```

To verify open a new terminal

For partitions:

```
hadoop fs -ls /user/hive/warehouse/shopping.db/
```

```
hadoop fs -ls /user/hive/warehouse/shopping.db/shopping3
```

You should see folders like:

```
category=bag  
category=cosmetic  
category=utensils  
category=apparel
```

For buckets:

```
hadoop fs -ls  
user/hive/warehouse/shopping.db/shopping3/category=utensils
```

You should see:

```
000000_0  
000001_0  
000002_0
```

🔗 That means **3 buckets created** ☒

2.

A. Implement Basic Data Handling Commands:

1. List the column names of DataFrame

```
import pandas as pd  
  
df = pd.read_csv("data.csv")  
  
# Column names  
  
print(df.columns)
```

2. Change column names of DataFrame

```
df.rename(columns={'old':'new'})
```

3. Bind sets of rows of data frames

```
df1=df2=df  
  
df3=pd.concat([df1,df2]).reset_index(drop=True)  
  
df3
```

4. Bind sets of columns of data frames

```
df1=df2=df  
  
df3=pd.concat([df1,df2],axis=1).reset_index(drop=True)  
  
df3
```

5. Find missing values in the dataset

```
df.isnull()
```

B. Implement Feature Extraction:

(Use packages that are applicable) using **Feature Subset Selection**
(Visualize Correlation using Heatmap)

```
import pandas as pd  
  
import seaborn as sns  
  
import matplotlib.pyplot as plt  
  
iris = pd.read_csv(r"C:\Users\DELL\Downloads\de lab prep\Iris1.csv")  
  
# Remove non-numeric columns automatically  
  
iris_num = iris.select_dtypes(include=['number'])  
  
# Correlation  
  
corr = iris_num.corr(method='pearson')
```

```
# Heatmap

sns.heatmap(corr, annot=True)

plt.show()


# Feature selection

selected_features = corr.columns[(corr.abs() > 0.5).any()]

print("Selected Features:", selected_features)
```

3.

A. Implement Basic Plots for Data Exploration (Use Iris dataset):

1. Generate box plot for each of the four predictors
2. Generate box plot for a specific feature
3. Generate histogram for a specific feature
4. Generate scatter plot of **petal length vs sepal length**

Check week 3

B. Implement Feature Extraction:

(Use packages that are applicable) using **Linear Discriminant Analysis (LDA)**

```
import pandas as pd

import matplotlib.pyplot as plt

from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

from sklearn.datasets import load_iris

# Load dataset

data = load_iris()
```

```
# Features and target
X = data.data
y = data.target

# Apply LDA
lda = LinearDiscriminantAnalysis(n_components=2)
X_lda = lda.fit_transform(X, y)

# Scatter plot
plt.scatter(X_lda[:, 0], X_lda[:, 1])
plt.xlabel("LD1")
plt.ylabel("LD2")
plt.title("LDA Scatter Plot")
plt.show()
```

if he asks for variance ratio add this line: `lda.explained_variance_ratio_`

4.

Implement Feature Construction:

(Use packages that are applicable) (**Ref: homeprices dataset**)

1. Dummy coding categorical (nominal) variables
2. Encoding categorical (ordinal) variables
3. Transforming numeric (continuous) features to categorical features

Entire week 5

5.

Implement Data Pre-Processing Methods:

Use **AutoMPG Dataset** and perform the following tasks:

1. Removing outliers / missing values using:
 - Percentiles
 - Standard Deviation
 - IQR
 - Z Score methods
2. Imputing standard values
3. Capping of values

```
import pandas as pd
import numpy as np
df = pd.read_csv("auto-mpg.csv")
print(df.head())
```

need to see this

6.

Implement Feature Extraction:

(Use packages that are applicable) with **PCA, LDA and SVD**

PCA

```
import pandas as pd
from sklearn.decomposition import PCA
from sklearn.datasets import load_iris
import matplotlib.pyplot as plt
```

```
# Load dataset
data = load_iris()
X = data.data
y = data.target
```

```
# Apply PCA
```

```

pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

# Display result
print(X_pca)

# Scatter plot
plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y)
plt.xlabel("PC1")
plt.ylabel("PC2")
plt.title("PCA Scatter Plot")
plt.show()

# Variance ratio
pca.explained_variance_ratio_

#covariance matrix
cov_mat=pca.get_covariance()
cov_mat

#eigen values
import numpy as np
eig_vals, eig_vecs=np.linalg.eig(cov_mat)
print("Eigenvelues \n%s" %eig_vals)
print("Eigenvectors \n%s" %eig_vecs)

```

LDA is in Q3

SVD

#Before SVD scatter plot

```

from sklearn.datasets import load_iris
iris1 = load_iris()
x=iris1.data
y=iris1.target
print(x.shape,y.shape)
#scatter plot
plt.scatter(x[:,0],x[:,1],c=y)

```

#After SVD scatter plot

```

from sklearn.decomposition import TruncatedSVD

```

```

# Apply SVD
svd = TruncatedSVD(n_components=2)
X_svd = svd.fit_transform(X)

```



```
# Display result
print(X_svd)

# Scatter plot
plt.scatter(X_svd[:, 0], X_svd[:, 1], c=y)
plt.xlabel("Component 1")
plt.ylabel("Component 2")
plt.title("SVD Scatter Plot")
plt.show()
```

7.

A. HDFS:

In vm
open terminal type `hadoop version`

Week 7 notes

- **Creating a file in local environment**

`gedit first.txt`

type some random text and click save then get back to the previous terminal in which u typed Hadoop version

type `ls` (to check if file is saved)

- **Display content of file**

`cat first.txt`

- **Move to Hadoop environment**

First we create a directory:
`hdfs dfs -mkdir /user/cse`

then:
`hdfs dfs -put first.txt /user/cse`

`hdfs dfs -ls /user/cse/`

extra just in case:

Hadoop to local sys

```
rm first.txt
```

```
hdfs dfs -copyToLocal /user/cse/first.txt
```

```
ls
```

create empty file in hdfs sys

```
hdfs dfs -touchz /user/cse/second.txt
```

```
hdfs dfs -ls
```

B. Implement Basic Statistical Functions for Data Exploration:

(Own choice of dataset)

1. Measures of Central Tendency: Mean, Median and Mode
2. Dispersion of data – Variance, Standard Deviation
3. Position of the different data values – Quartiles, Interquartile Range (IQR)

```
import pandas as pd
df = pd.read_csv(r"C:\Users\DELL\Downloads\auto-mpg.csv")
```

```
# Select only numeric columns (important)
df = df.select_dtypes(include=['number'])
df = df.drop(['Unnamed: 0', 'Unnamed: 0.1'], axis=1)
```

```
# -----
# 1. Central Tendency
# -----
print("Mean:\n", df.mean())
print("\nMedian:\n", df.median())
print("\nMode:\n", df.mode().iloc[0])
```

```
# -----
# 2. Dispersion
# -----
```

```

print("\nVariance:\n", df.var())
print("\nStandard Deviation:\n", df.std())

# -----
# 3. Position
# -----
print("\nQuartiles:\n", df.quantile([0.25, 0.5, 0.75]))

# IQR
Q1 = df.quantile(0.25)
Q3 = df.quantile(0.75)
IQR = Q3 - Q1

print("\nInterquartile Range (IQR):\n", IQR)

```

8.

Develop **MapReduce example program** in a MapReduce environment to find out the **number of occurrences of each word in a text file**.

9.

A.

Create a database **shopping**, create table (**shopping1**) inside the database shopping,
Load the data in HIVE table from local

(Suppose you have a file shop.txt in your local machine, you can create a CSV file using gedit command)

1. Open terminal in the vm
2. Type hive
3. create database shopping; use shopping;
4. create table shopping1(
 - code int,
 - item_name string,
 - category string,
 - place string
)
 - row format delimited fields terminated by ',';

in a new terminal type

gedit shop.txt

in that type randome stuff like:

- 1,purse,bag,shimla
- 2,lipstick,cosmetic,delhi
- 3,bowl,utensils,jammu
- 4,mobile,electronicgadget,hyderabad
- 5,skirt,apparel,chennai
- 6,bedcover,furnishing,chandigarh
- 7,car,toys,karnal
- 8,handpurse,bag,solan
- 9,cream,cosmetic,jodhpur
- 10,plate,utensils,Mohali

And then save it

Get back to the hive terminal

```
LOAD DATA LOCAL INPATH '/home/cloudera/Desktop/shop.txt'  
OVERWRITE INTO TABLE shopping1;
```

Note: incase it says file missing or smtg then the path must be wrong
In a different terminal **type find /home -name "shop.txt"** to get correct path.

```
select * from shopping1;
```

B. Implement Feature Extraction:

(Use packages that are applicable) using **Principal Component Analysis (PCA)**

10.

Create a database and work with **MongoDB Query Language**

(Ex: Book / Any Choice Database)

- a. Creating a Database and Collection
- b. Insert Documents: Single insert, Multiple inserts
- c. Find All Books & Find with Condition
- d. Update One Book & Multiple Books
- e. Delete Documents (One & Multiple Books)